A REDUCED SEPARATED SCHEME OF FINITE TYPE OF DIMENSION AT MOST ONE ADMITS AN H-PROJECTIVE COMPLETION WITH DISCRETE VALUATION RING BOUNDARY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Projective completion of a separated scheme of finite type of dimension at most one. Stacks Project, Varieties, Lemma lemma-dim-1-projective-completion (tag 0A26). Produces the initial immersion X -> Xbar that is then normalized.
- Relative normalization is finite and normal schemes are discrete valuation rings in codimension one. Stacks Project, Morphisms, Lemma morphisms-lemma-relative-normalization-finite (tag 0BXL) and Lemma morphisms-lemma-relative-normalization-normal-codim-1 (tag 0BXN). The first gives finiteness of the normalization and the second identifies the boundary local rings as discrete valuation rings.

Source reference. Original lemma 0BXV: sources/stacks-chow/tags.txt, source lines 14993–14993. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BXS: sources/stacks-chow/tags.txt, source lines 14990–14990. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B3I: sources/stacks-chow/tags.txt, source lines 13966–13966. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 087S: sources/stacks-chow/tags.txt, source lines 10437–10437. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01WE: sources/stacks-chow/tags.txt, source lines 2479–2479. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0BXB: sources/stacks-chow/tags.txt, source lines 14974–14974. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a separated scheme of finite type over k. If X is reduced and $\dim(X) \leq 1$, then there exists an open immersion $j : X \to \overline{X}$ such that*

(1) *$\overline{X}$ is H-projective over k, i.e., $\overline{X}$ is a closed subscheme of $\mathbf{P}^d_k$ for some d,*
(2) *$j(X) \subset \overline{X}$ is dense and scheme theoretically dense,*
(3) *$\overline{X} \setminus X = \{x_1, \ldots, x_n\}$ for some closed points $x_i \in \overline{X}$,*
(4) *the local rings $\mathcal{O}_{\overline{X}, x_i}$ are discrete valuation rings for $i = 1, \ldots, n$.*

Proof. Let $j : X \to \overline{X}$ be as in Lemma 0BXV. Consider the normalization X' of $\overline{X}$ in X. By Lemma 0BXS the morphism $X' \to \overline{X}$ is finite. By Morphisms, Lemma 0B3I $X' \to \overline{X}$ is projective. By Morphisms, Lemma 087S we see that $X' \to \overline{X}$ is H-projective. By Morphisms, Lemma 01WE we see that $X' \to \mathrm{Spec}(k)$ is H-projective. Let $\{x'_1, \ldots, x'_m\} \subset X'$ be the inverse image of $\{x_1, \ldots, x_n\} = \overline{X} \setminus X$. Then $\dim(\mathcal{O}_{X', x'_i}) = 1$ for all $1 \leq i \leq m$. Hence the local rings $\mathcal{O}_{X', x'}$ are discrete valuation rings by Morphisms, Lemma 0BXB. Then $X \to X'$ and $\{x'_1, \ldots, x'_m\}$ is as desired. □